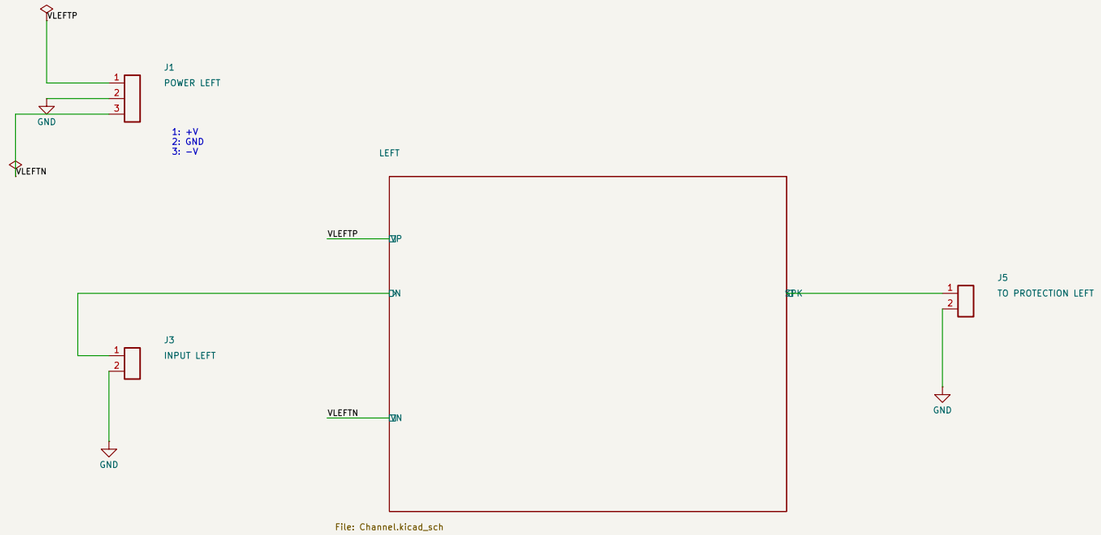
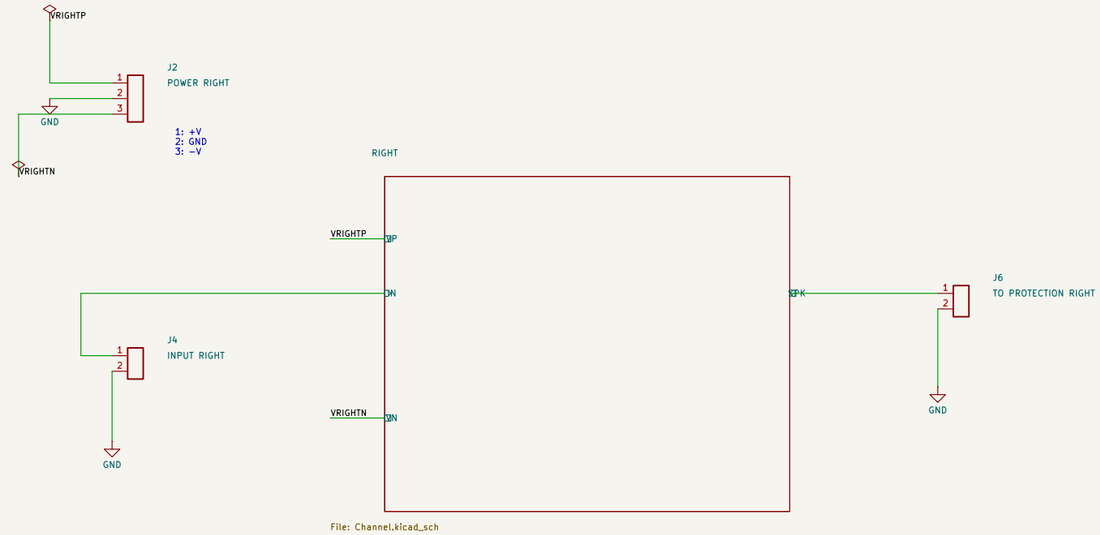
# MOSFETAmp2 – STEREO DISCRETE CLASS-AB

REV 0.2: SIMULATION DEVELOPMENT / NOT FABRICATION RELEASED
8 ohm prototype; external current-limited +/-12 to +/-15 V first.
No mains circuitry. External DC speaker protection required. Read README and bench procedure.
BIAS: fixed-ratio sensor + programmed offset; MOSFET VGS slope and SOA still need measurement.




Assembly gates
1. Fit bias short J101/J201; leave R114/R214 unfitted; trimmers at MAXIMUM resistance.
2. Verify live tab insulation and Q108/Q208 thermal contact.
3. Connect supply GND to the junction of the two series bench outputs.
4. Connect oscilloscope earth clips to board GND only.
5. Use dummy loads; no speakers until DC protection and warm-up tests pass.

GND

External PSU assumption envelope: roughly +/-12.5 V valley to +/-20.6 V peak.
Final transformer and capacitors not fixed. Recalculate and measure before use.
Schematic connected and ERC checked separately from electrical/model/thermal validation.
The original empty PCB is retained. Layout is for the user; no routing supplied.

| | | |
|---|---|---|
| Sheet: / | | |
| File: MOSFETAmp2.kicad_sch | | |
| ***Title: MOSFETAmp2 stereo Class-AB*** | | |
| Size: A3 | Date: 2026-10-25 | **Rev: 0.2 EXPERIMENTAL** |
| KiCad E.D.A. 10.0.5 | | Id: 1/3 |

INPUT / DIFFERENTIAL PAIR + MIRROR

BUFFERED VAS / DOMINANT POLE

DRIVERS / ONE COMPLEMENTARY PAIR

VP
D102
1N4148
D101
1N4148
R108
12k
VN
R101
1k
GIN
B1
R102
22k
C101
220p C0G
R103
22k
OUT
GND
FB
GND
R104
2k2
GND
R107
330R
Q105
BC556
TREF
R105
100R
R106
100R
Q101
BC556
Q102
BC556
C2N
Q103
BC546
Q104
BC546
VN
D105
1N4148
FB
B1
D106
1N4148

Q101/Q102: match at 1 mA; bond flat faces.
R101 + R102: input about 23k.
Gain = 11 from B1: about 10.52 at IN.
D105/D106 limit differential overload.

VP
C102
C2N
BN
330p C0G
D107
1N4148
D108
BAT46
C2N
BN
Q115
BC546
C2N
BN
Q106
BD139
R110
10k
R109
100R
VN
VN

C102 = 330p C0G, 50V, radial 2.54mm.
D107 + D108 prevent VAS saturation.
No rail bootstrapping or regulators.
VAS/load current about 6 mA.

VP
R111
100R
TREF
Q107
BC556
BP

VP
BP
Q109
BD139
R117
100R
GN
Q111
IRF540N
R115
2k2
OUT
R116
2k2
SN
R119
0R33 / 5W
TP101
SN
SN
OUT
L101
0.7uH AIR / 3A
SPK
R120
0R33 / 5W
R126
10R / 2W
SP
TP103
SP
Q112
IRF9540N
SP
R118
100R
GP
Q110
BD140
BN
VN
VN
R125
10R / 2W
C104
100n FILM / 63V
GND

Local bypass:
220u / 35V each rail
100n / 63V each rail
C106 positive to GND

Measure 16.5 mV per 0R33 for 50 mA.
OUT is feedback sense BEFORE the coil.
MOSFET drain tabs are live: INSULATE.
8 ohm prototype: no short-circuit protection.

VP
TP104
VP
TP105
VN
VP
C105
220u / 35V
GND
C106
220u / 35V
VN
VP
C107
100n / 63V
GND
TP102
OUT
OUT
C108
100n / 63V
VN
VN

OPERATING LIMITS

8 ohm prototype: current-limited bench PSU first.
No output short-circuit protection is fitted.
External DC speaker protection is required.
Gate clamps protect VGS; they do not establish SOA.
Read the bias and warm-up procedure before power-up.

THERMAL BIAS / PROGRAMMED OFFSET

J101
BIAS SHORT - FIT FIRST
BP
BN
VP
R127
680R
TREF
Q117
BC556
BREF
Q116
BC546
D113
LED RED
R114
1k SOT
RV101
200R MULTITURN
VN
BP
D109
1N4148
D112
1N4148
R112
3k9
R113
1k SLOPE SEL
BN
Q108
BD139
BP
C103
100n / 63V
BN

Q108 on heatsink beside MOSFETs: ratio fixed by R112/R113 (R113 = slope select).
D109/D112 track the drivers on the PCB.
Offset = 3k9 x (V(D113) - VBE)/(R114 + RV101).
R114 unfitted = zero offset = minimum bias.
RV101 maximum resistance = minimum bias.

GN
R121
47k
SN
GN
D110
BZX85C10
SN
SP
R122
47k
GP
SP
D111
BZX85C10
GP

Gate-source Zeners
N: K to gate, A to source
P: K to source, A to gate

VN

Sheet: /LEFT/
File: Channel.kicad_sch

**Title: Class-AB channel - experimental Rev 0.2**

Size: A3 | Date: 2026-10-25 | **Rev: 0.2 EXPERIMENTAL**
KiCad E.D.A. 10.0.5 | Id: 2/3

INPUT / DIFFERENTIAL PAIR + MIRROR

BUFFERED VAS / DOMINANT POLE

DRIVERS / ONE COMPLEMENTARY PAIR

VP
D202
1N4148
D201
1N4148
R208
12k
VN
R207
330R
TREF
Q205
BC556
R201
1k
GIN
B1
R202
22k
C201
220p C0G
R205
100R
R206
100R
R203
22k
OUT
GND
FB
GND
Q201
BC556
Q202
BC556
C2N
R204
2k2
GND
Q203
BC546
Q204
BC546
VN
D205
1N4148
FB
B1
D206
1N4148
Q101/Q102: match at 1 mA; bond flat faces.
R101 + R102: input about 23k.
Gain = 11 from B1: about 10.52 at IN.
D105/D106 limit differential overload.
VP
C202
C2N
BN
330p C0G
D207
1N4148
D208
BAT46
C2N
BN
Q215
BC546
C2N
BN
Q206
BD139
R210
10k
R209
100R
VN
VN
C102 = 330p C0G, 50V, radial 2.54mm.
D107 + D108 prevent VAS saturation.
No rail bootstrapping or regulators.
VAS/load current about 6 mA.
VP
R211
100R
TREF
Q207
BC556
BP
VP
BP
Q209
BD139
R217
100R
GN
Q211
IRF540N
VP
R215
2k2
OUT
R216
2k2
R218
100R
GP
Q212
IRF9540N
BN
Q210
BD140
VN
VN
SN
R219
0R33 / 5W
OUT
SP
R220
0R33 / 5W
SP
TP204
VP
TP205
VN
VP
VP
C205
220u / 35V
GND
GND
C206
220u / 35V
VN
VP
C207
100n / 63V
GND
GND
C208
100n / 63V
VN
TP201
SN
SN
TP202
OUT
OUT
L201
0.7uH AIR / 3A
SPK
R226
10R / 2W
TP203
SP
R225
10R / 2W
C204
100n FILM / 63V
GND
Local bypass:
220u / 35V each rail
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BN
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R227
680R
TREF
Q217
BC556
BREF
Q216
BC546
D213
LED RED
R214
1k SOT
RV201
200R MULTITURN
VN
BP
D209
1N4148
D212
1N4148
R212
3k9
R213
1k SLOPE SEL
BN
Q208
BD139
C203
100n / 63V
BN
Q108 on heatsink beside MOSFETs: ratio fixed by
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R114 unfitted = zero offset = minimum bias.
RV101 maximum resistance = minimum bias.
VN
GN
R221
47k
SN
GN
D210
BZX85C10
SN
SP
R222
47k
GP
SP
D211
BZX85C10
GP
Gate-source Zeners
N: K to gate, A to source
P: K to source, A to gate
Sheet: /RIGHT/
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**Title: Class-AB channel - experimental Rev 0.2**

Size: A3 | Date: 2026-10-25 | **Rev: 0.2 EXPERIMENTAL**

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